

Confidence Intervals

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Topics covered:

- ① Definition of a confidence interval
- ② Confidence interval for the mean
 - Known variance (unrealistic)
 - Unknown variance
- ③ Confidence interval for a proportion
- ④ Sample size calculation
 - Infinite population
 - Finite population

Confidence interval:

- A range of values based on a sample such that the population parameter, i.e., mean or proportion, is occurring in the range with probability α . The specific probability is called level of confidence and is in most cases set to 95%.
- Confidence interval = point estimate $\pm$ margin of error

Put differently, if you take 100 samples and construct a confidence interval as outlined in the next slides; in 95 cases, that confidence interval will contain the population mean or proportion. The confidence interval is influenced by

- Sample size n
- Population standard deviation σ (often estimated by s)
- Level of confidence

Confidence Interval for the Mean

Components for any confidence interval involving the mean:

- Sample size
- Sample mean

Components depending on what we know about the population standard deviation:

- Known population standard deviation
 - Population standard deviation: σ
 - z-value for a given confidence level
- Unknown population standard deviation
 - Estimate of the population standard deviation: s
 - t-value for a given confidence level

In what follows, only a confidence interval for the mean with *unknown* variance is considered.

Confidence Interval for the Mean: Overview

Given that the standard deviation of the population is unknown, the confidence interval is constructed as follows:

$$\bar{x} \pm t_{\alpha, df} \cdot \frac{s}{\sqrt{n}}$$

Notes:

- Requires estimation of the population variance s^2 and standard deviation through s .
- Use of the t -distribution instead of the standard normal distribution:
 - α is the confidence level, e.g., 95%
 - df are the degrees of freedom

Confidence Interval for the Mean: Steps

Steps to construct the confidence interval if σ is unknown:

- ① Estimate the sample mean $\bar{x}$
- ② Determine the degrees of freedom $df = n - 1$
- ③ Determine $t_{\alpha, df}$: For a 95% confidence interval, this can be done with any statistical software:
 - In R: `qt(0.975, df=N-1)`
- ④ Use the equation:

$$\bar{x} \pm t_{\alpha, df} \cdot \frac{s}{\sqrt{n}}$$

Confidence Interval for the Mean: Example

Assume the American Economic Association (AEA) wants to construct a 95% confidence interval for the starting salaries of economics majors. The sample mean of 36 randomly selected graduates is \$48,500. The calculated sample variance is $s = \$3,600$.

$$\$48,500 \pm 2.03 \cdot \frac{\$3,600}{\sqrt{36}} = \$48,500 \pm \$1,218$$

The value of 2.03 leaves 2.5% in each tail of the t -distribution with 35 degrees of freedom. The \$1,218 represents the margin of error. The value of \$600 (i.e., $\$3,600 = / \sqrt{36}$) represents the standard error.

Confidence Interval for the Mean: R

Confidence Interval for the Mean

```
nobs = nrow(mh2)
meandata = mean(mh2$price)
stdev = sd(mh2$price)
t_alpha_df = qt(0.975,nobs-1)
CI_lower = meandata-t_alpha_df*stdev/sqrt(nobs)
CI_upper = meandata+t_alpha_df*stdev/sqrt(nobs)
t.test(mh2$price)
```

Confidence Interval for a Proportion: Overview

For a proportion, we have the following:

- Estimate proportion from the data: $\hat{p}$
- Estimate standard deviation: $\sigma = \sqrt{\hat{p} \cdot (1 - \hat{p})}$

So the standard error for the proportion is

$$\sigma_{\hat{p}} = \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}}$$

Thus, the 95% confidence interval is constructed as follows:

$$\hat{p} \pm 1.96 \cdot \sigma_{\hat{p}} \Leftrightarrow \hat{p} \pm 1.96 \cdot \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}}$$

Confidence Interval for a Proportion: Example

Illustration

Assume that you are interested in the political party affiliation of voters. Suppose that $n = 1000$ and $p_{GOP} = 0.55$. Given the equation on the previous slide, you can calculate:

$$\sigma_{\hat{p}} = \sqrt{\frac{0.55 \cdot (1 - 0.55)}{1000}} = 0.0157$$

This gives us the margin of error for the political party affiliation of $0.55 \pm 1.96 \cdot 0.0157$.

Confidence Interval for a Proportion: R

Illustration

```
vote20      = subset(gss,vote20 %in% c(1,2),  
                    select=c("year","vote20"))  
vote20      = ifelse(vote20$vote20==1,1,0)  
nobs        = length(vote20)  
meandata    = mean(vote20)  
z           = qnorm(0.975)  
stderror    = sqrt(meandata*(1-meandata)/nobs)  
CI_lower    = meandata-z*stderror  
CI_upper    = meandata+z*stderror  
t.test(vote20)
```

Sample Size Calculations

Recall from the confidence interval that

$$p = z \cdot \frac{\sigma}{\sqrt{n}}$$

Margin of error depends on sample size n . Possibility of calculation the sample size necessary to achieve a given margin of error. Two possible cases:

- Infinite population
- Finite population

Equation:

$$z \cdot \sqrt{\frac{\sigma^2}{n}} \leq \epsilon \quad \Rightarrow \quad n \geq \left(\frac{z \cdot \sigma}{\epsilon} \right)^2$$

Sample Size Calculations: Infinite Population

With prior knowledge about the proportion

$$n \geq \frac{z^2 \cdot p \cdot (1 - p)}{\epsilon^2}$$

Without prior knowledge about the proportion

$$n \geq \left(\frac{z \cdot 0.5}{\epsilon} \right)^2$$

Sample Size Calculations: Example

Suppose you want to know how many people support a property tax reform. You do not have any knowledge about the population parameters but want the estimate to be within 2%. For this reason, you adopt an initial estimate of $p = 0.5$. This results in a “worst case” scenario.

$$n = \left(\frac{1.96 \cdot 0.5}{0.02} \right)^2 = 2401$$

Sample Size Calculations: Finite Population

The sample size necessary also depends on the population size. Suppose you are interested in how many students support a privatization of parking.

$$n_f = \frac{n_\infty \cdot N}{n_\infty + (N - 1)}$$

For a college with 10,000 students:

$$\frac{2401 \cdot 10000}{2401 + (10000 - 1)} = 1937$$

The sample size needed for large-sample confidence intervals is $n \cdot \hat{p} \geq 15$. Sometimes it is easier to go with a different rule thumb that specifies that $n \geq 30$.